gps_comparison

April 6, 2026

1 GPS Comparison

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Relevant Links: - Phone GPS Logger Monday Update - Phone GPS Logger Project - Brightside Driving Day Note-Taking Sheet

The purpose of this project is to take an initial look at the GPS data logged by Brightside's TEL board in our April 2nd driving day at the McArthur Glen Designer Outlet parking lot and compare it to the data collected by my Redmi Note 10 Pro (CN) running the GPSLogger app.

1.1 Imports

Import any packages or dependencies for this project. Uncomment or delete these lines as needed, and add other dependencies

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
from data_tools import query, TimeSeries
from datetime import datetime
```

1.2 Data Acquisition

Query GPS and speed data from April 2nd driving day.

[2]:
```
# 1. Define fields to query
gps_fields: list[str] = [
    "GPSLatitude", "GPSLongitude", "GPSSatCount", "GPSSNR"
]
# Ignore GPSLatSide and GPSLonSide which say if you're N or S of the equator␣
↪and E or W of the Grenwich Meridian
# We're in vancouver so we should leave the latitude sign positive and make␣
↪longitude negative.
other_fields: list[str] = [
    "MotorRotatingSpeed"
]
fields_to_query = gps_fields + other_fields
```

```
# 2. Define UTC time period
driving_day_start = datetime.fromisoformat("2026-04-02T17:00:00Z") # 9AM PDT
driving_day_end   = datetime.fromisoformat("2026-04-03T01:00:00Z") # 5PM PDT␣
↪(same day in Vancouver)

# 3. Query InfuxDB data
client = query.DBClient()
query_data: dict[str, TimeSeries] = {}
for field in fields_to_query:
    query_data[field] = client.query_time_series(
        driving_day_start,
        driving_day_end,
        field,
    )
```

[3]:
```
for k, v in query_data.items():
    print(f"{k:20}: start={v.start}, stop={v.stop}, num_points={len(v)}")
```

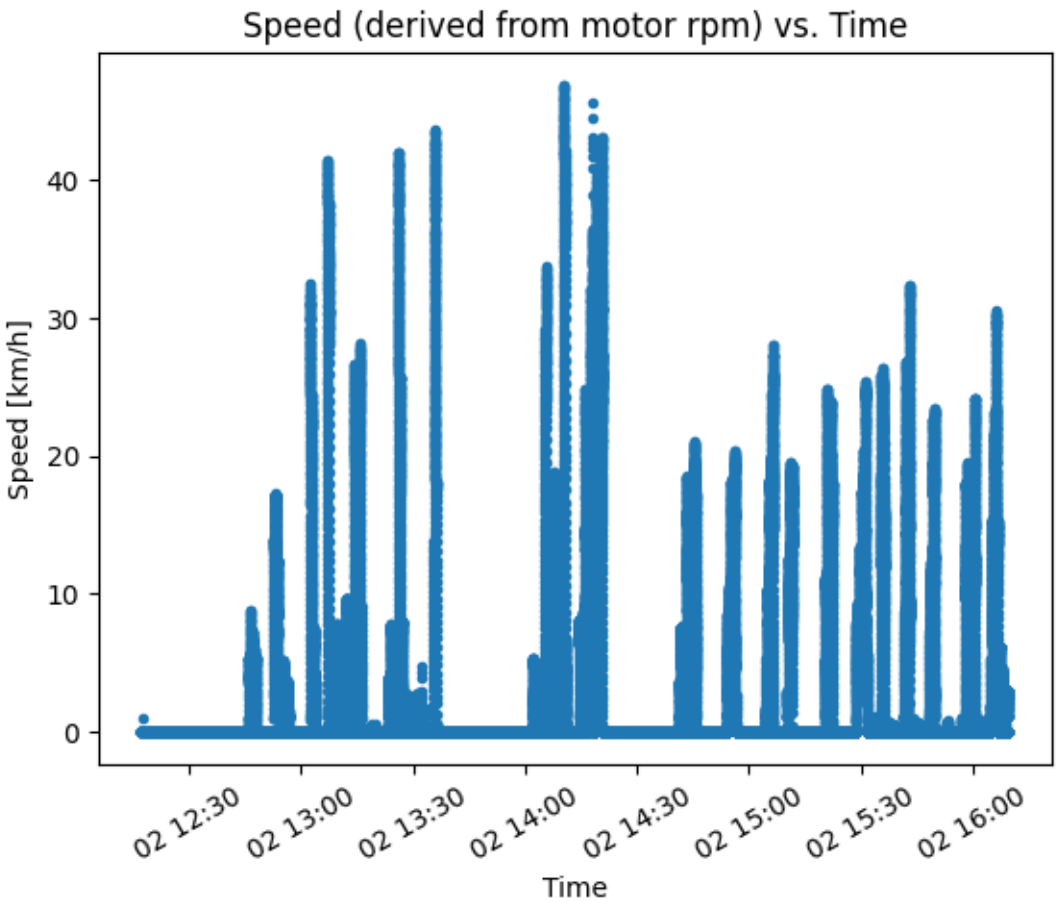
```
GPSLatitude          : start=2026-04-02 11:31:48.892000+00:00, stop=2026-04-02
16:09:23.178000+00:00, num_points=166543
GPSLongitude         : start=2026-04-02 11:31:48.892000+00:00, stop=2026-04-02
16:09:23.178000+00:00, num_points=166543
GPSSatCount          : start=2026-04-02 11:31:48.830000+00:00, stop=2026-04-02
16:09:23.128000+00:00, num_points=166543
GPSSNR               : start=2026-04-02 11:31:48.830000+00:00, stop=2026-04-02
16:09:23.128000+00:00, num_points=166543
MotorRotatingSpeed   : start=2026-04-02 12:17:23.395000+00:00, stop=2026-04-02
16:09:23.315000+00:00, num_points=139200
```

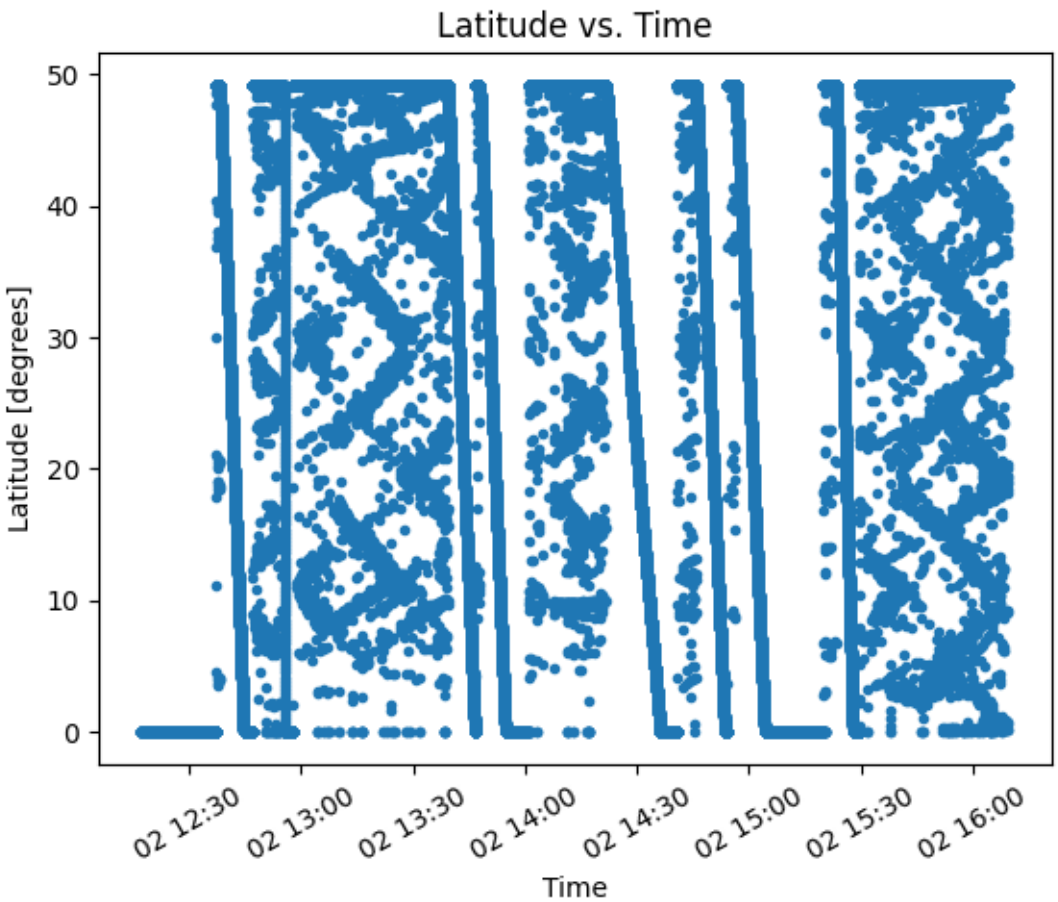
1.3 Analysis

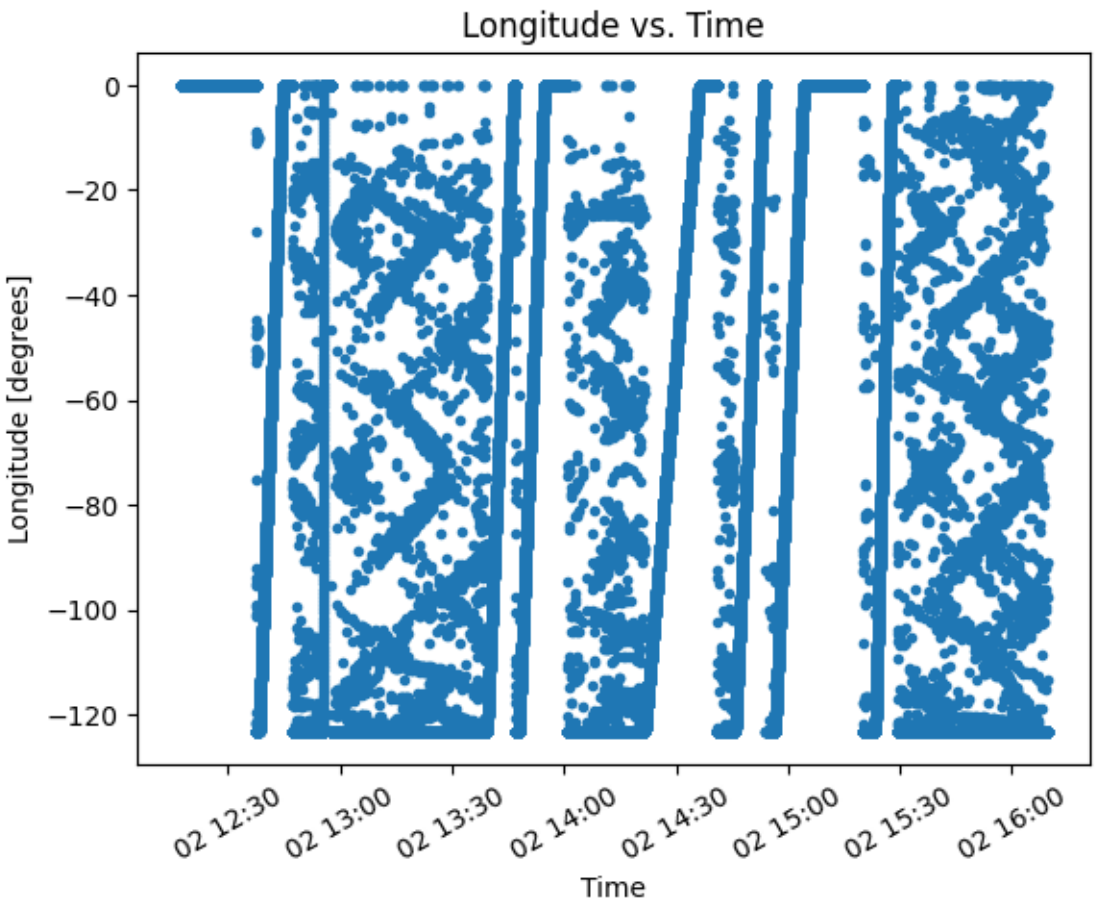
[4]:
```
latitude:  TimeSeries = None
longitude: TimeSeries = None
sat_count: TimeSeries = None
gps_snr:   TimeSeries = None
speed_kph: TimeSeries = None
latitude, longitude, sat_count, gps_snr, speed_kph = TimeSeries.align(
    *[query_data[field] for field in fields_to_query]
)

assert np.all(speed_kph.x_axis == latitude.x_axis)  # Quick alignment check
assert np.all(speed_kph.datetime_x_axis == latitude.datetime_x_axis)
t_axis = speed_kph.x_axis
dt_axis = speed_kph.datetime_x_axis

# Since we are in the western hemisphere, flip the sign on the (unsigned)␣
↪latitude value
```

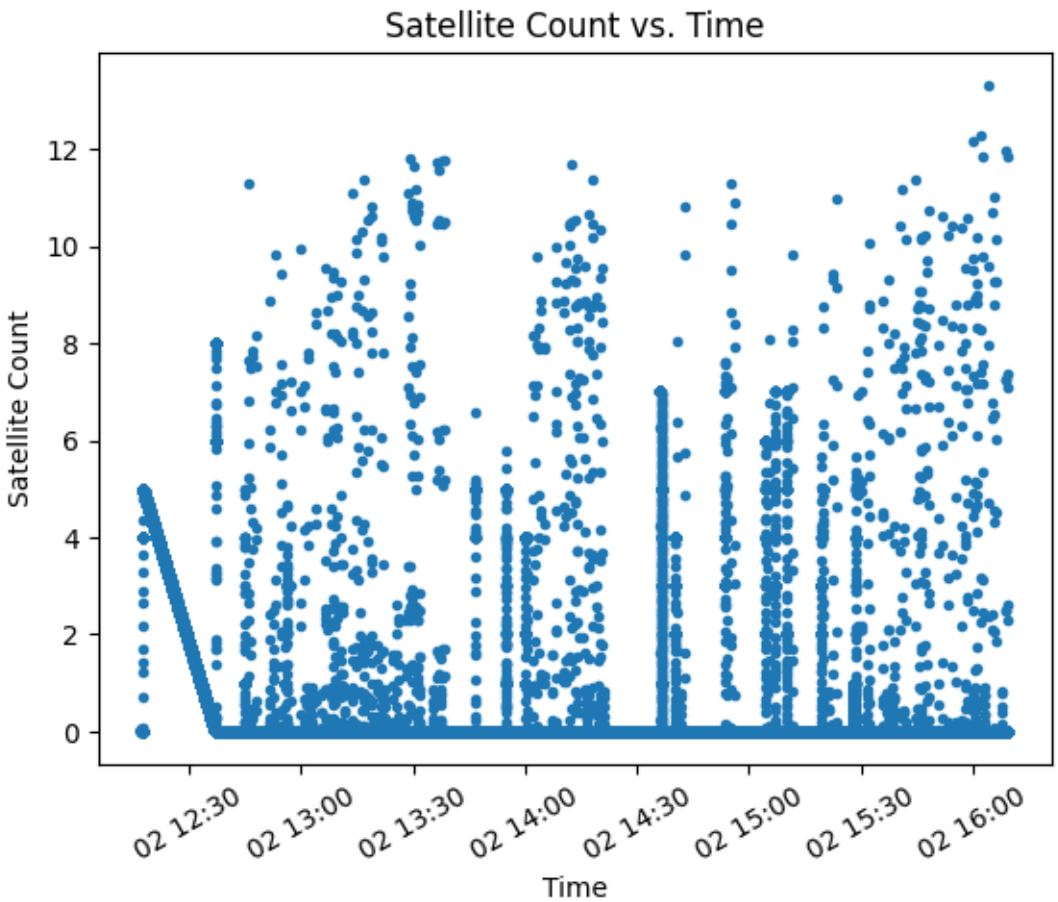
```
# We could use GPSLatSide to determine this but it's not necessary since I know␣
↪where we are
longitude = -longitude
```

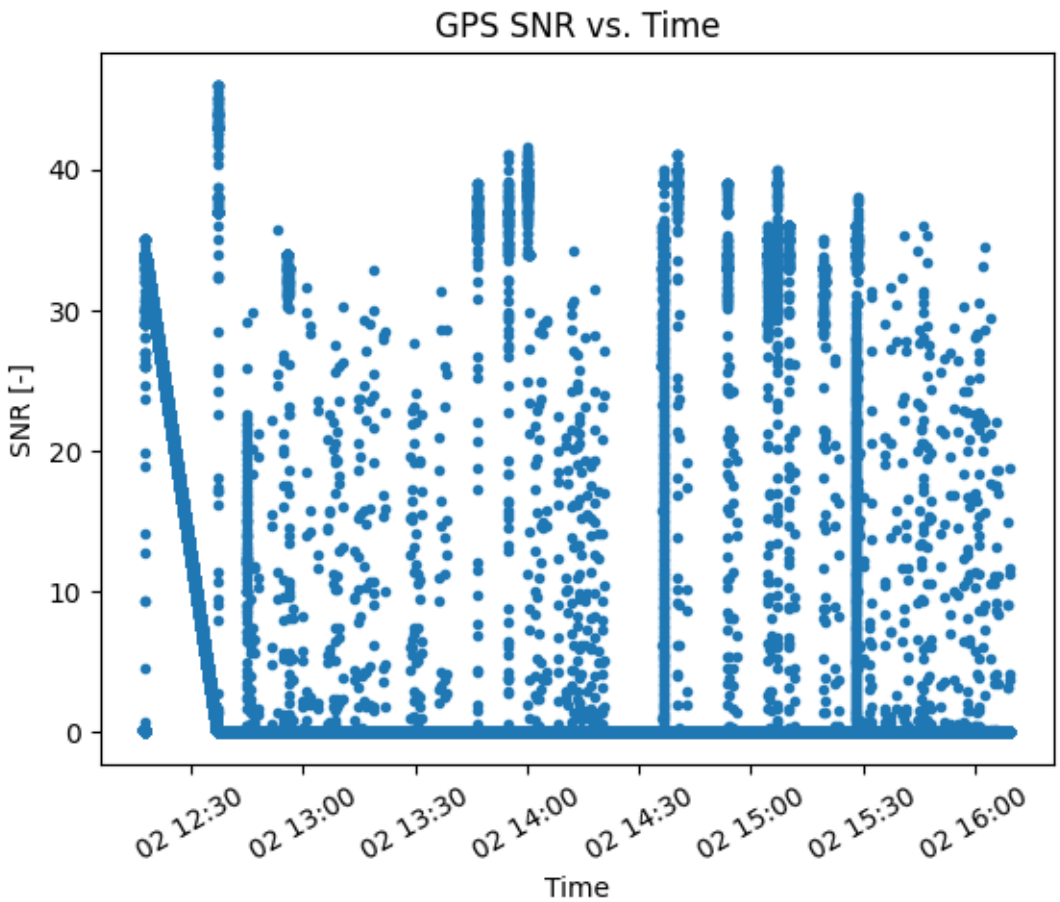
[5]:
```
plt.scatter(dt_axis, speed_kph, marker='.')
plt.title("Speed (derived from motor rpm) vs. Time")
plt.xlabel("Time")
plt.xticks(rotation=30)
plt.ylabel("Speed [km/h]")
plt.show()

plt.scatter(dt_axis, latitude, marker='.')
plt.title("Latitude vs. Time")
plt.xlabel("Time")
plt.xticks(rotation=30)
plt.ylabel("Latitude [degrees]")
plt.show()

plt.scatter(dt_axis, longitude, marker='.')
plt.title("Longitude vs. Time")
plt.xlabel("Time")
plt.xticks(rotation=30)
plt.ylabel("Longitude [degrees]")
plt.show()

plt.scatter(dt_axis, sat_count, marker='.')
plt.title("Satellite Count vs. Time")
plt.xlabel("Time")
plt.xticks(rotation=30)
plt.ylabel("Satellite Count")
plt.show()

plt.scatter(dt_axis, gps_snr, marker='.')
plt.title("GPS SNR vs. Time")
plt.xlabel("Time")
plt.xticks(rotation=30)
plt.ylabel("SNR [-]")
plt.show()
```


Speed (derived from motor rpm) vs. Time
Speed [km/h]
0
10
20
30
40
02 12:30
02 13:00
02 13:30
02 14:00
02 14:30
02 15:00
02 15:30
02 16:00
Time


Latitude vs. Time
Latitude [degrees]
50
40
30
20
10
0
02 12:30
02 13:00
02 13:30
02 14:00
02 14:30
02 15:00
02 15:30
02 16:00
Time


Longitude vs. Time
Longitude [degrees]
0
−20
−40
−60
−80
−100
−120
02 12:30
02 13:00
02 13:30
02 14:00
02 14:30
02 15:00
02 15:30
02 16:00
Time


Satellite Count vs. Time
Satellite Count
0
2
4
6
8
10
12
02 12:30
02 13:00
02 13:30
02 14:00
02 14:30
02 15:00
02 15:30
02 16:00
Time



1.4 Filtering the GPS data

The latitude and longitude data is sometimes zero, which results in messed up values during interpolation when converting to TimeSeries.

The solution is to query the raw dataframe, filter out zeros and then convert to TimeSeries manually.

[6]:
```
import pandas as pd


latitude_df: pd.DataFrame = client.query_series(
    driving_day_start,
    driving_day_end,
    "GPSLatitude",
)
longitude_df: pd.DataFrame = client.query_series(
    driving_day_start,
    driving_day_end,
    "GPSLongitude",
```

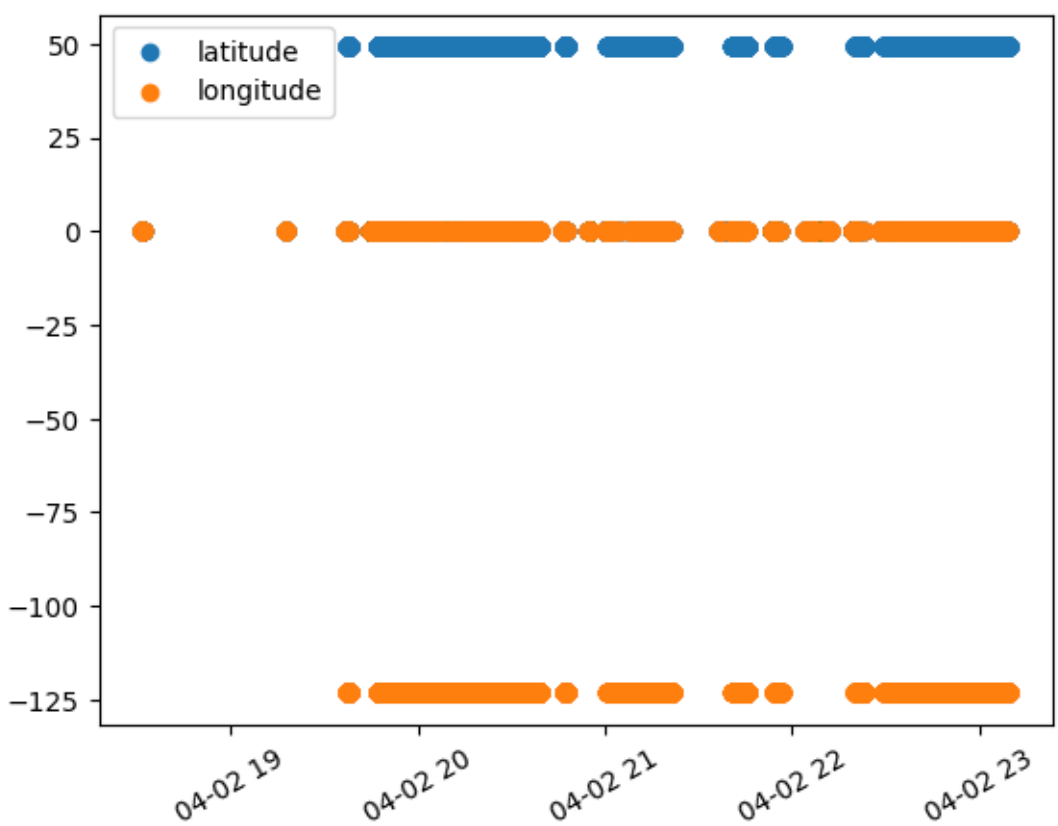
```
)

display(latitude_df)
display(longitude_df)
```

```
        result  table                     _start                       _stop  \
0      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
1      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
2      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
3      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
4      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
…        …    …                         …                          …
29779  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29780  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29781  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29782  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29783  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00

                                _time _measurement         car        class  \
0     2026-04-02 11:31:48.892000+00:00          TEL  Brightside  GPSPosition
1     2026-04-02 11:31:49.145000+00:00          TEL  Brightside  GPSPosition
2     2026-04-02 11:31:49.397000+00:00          TEL  Brightside  GPSPosition
3     2026-04-02 11:31:49.650000+00:00          TEL  Brightside  GPSPosition
4     2026-04-02 11:31:49.902000+00:00          TEL  Brightside  GPSPosition
…                                  …           …          …           …
29779 2026-04-02 16:09:21.877000+00:00          TEL  Brightside  GPSPosition
29780 2026-04-02 16:09:22.421000+00:00          TEL  Brightside  GPSPosition
29781 2026-04-02 16:09:22.672000+00:00          TEL  Brightside  GPSPosition
29782 2026-04-02 16:09:22.926000+00:00          TEL  Brightside  GPSPosition
29783 2026-04-02 16:09:23.178000+00:00          TEL  Brightside  GPSPosition

       GPSLatitude
0         0.000000
1         0.000000
2         0.000000
3         0.000000
4         0.000000
…            …
29779     0.000000
29780    49.198734
29781    49.198734
29782    49.198734
29783    49.198734

[29784 rows x 9 columns]
        result  table                     _start                       _stop  \
0      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
```

```
1      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
2      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
3      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
4      _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
…        …    …                        …                        …
29779  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29780  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29781  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29782  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00
29783  _result      0 2026-04-02 10:00:00+00:00 2026-04-02 18:00:00+00:00

                                _time _measurement         car        class  \
0     2026-04-02 11:31:48.892000+00:00          TEL  Brightside  GPSPosition
1     2026-04-02 11:31:49.145000+00:00          TEL  Brightside  GPSPosition
2     2026-04-02 11:31:49.397000+00:00          TEL  Brightside  GPSPosition
3     2026-04-02 11:31:49.650000+00:00          TEL  Brightside  GPSPosition
4     2026-04-02 11:31:49.902000+00:00          TEL  Brightside  GPSPosition
…                                  …           …          …           …
29779 2026-04-02 16:09:21.877000+00:00          TEL  Brightside  GPSPosition
29780 2026-04-02 16:09:22.421000+00:00          TEL  Brightside  GPSPosition
29781 2026-04-02 16:09:22.672000+00:00          TEL  Brightside  GPSPosition
29782 2026-04-02 16:09:22.926000+00:00          TEL  Brightside  GPSPosition
29783 2026-04-02 16:09:23.178000+00:00          TEL  Brightside  GPSPosition

       GPSLongitude
0          0.000000
1          0.000000
2          0.000000
3          0.000000
4          0.000000
…             …
29779      0.000000
29780    123.145302
29781    123.145302
29782    123.145302
29783    123.145302

[29784 rows x 9 columns]
```

[7]:
```
# Ensure they have the same time axis
assert np.all(latitude_df["_time"] == longitude_df["_time"])

latitude_df_dtxaxis = latitude_df.set_index(
    pd.to_datetime(latitude_df["_time"]).dt.tz_localize(None).dt.
 ↪tz_localize('Etc/GMT+7').dt.tz_convert('UTC')
)
longitude_df_dtxaxis = longitude_df.set_index(
```

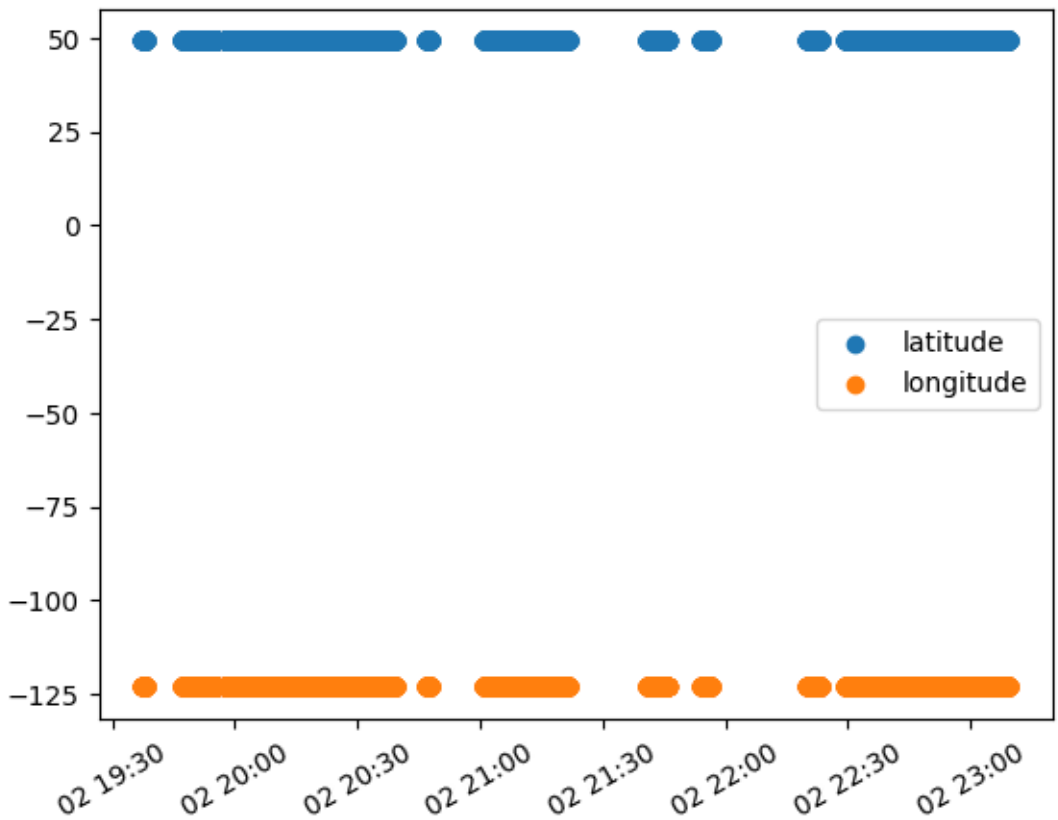
```
    pd.to_datetime(longitude_df["_time"]).dt.tz_localize(None).dt.
 ↪tz_localize('Etc/GMT+7').dt.tz_convert('UTC')
)

gps_latlon_df_raw = latitude_df_dtxaxis[["GPSLatitude"]].
 ↪join(longitude_df_dtxaxis[["GPSLongitude"]], how='inner')
gps_latlon_df_raw["GPSLongitude"] = -gps_latlon_df_raw["GPSLongitude"]

fig, ax = plt.subplots()
ax.scatter(gps_latlon_df_raw.index, gps_latlon_df_raw['GPSLatitude'],␣
 ↪label='latitude')
ax.scatter(gps_latlon_df_raw.index, gps_latlon_df_raw['GPSLongitude'],␣
 ↪label='longitude')
plt.xticks(rotation=30)
plt.legend(loc='best')
plt.show()
```



We can see above that the values are now either zero or in the correct range. We can also see that they are zero at the same time, i.e., when we have no fix.

Given this, the filtering is straight forward.

[8]:
```
gps_latlon_df = gps_latlon_df_raw[(gps_latlon_df_raw['GPSLatitude'].abs() >= 0.
↪1)
                                  & (gps_latlon_df_raw['GPSLongitude'].
↪abs() >= 0.1)]

fig, ax = plt.subplots()
ax.scatter(gps_latlon_df.index, gps_latlon_df['GPSLatitude'], label='latitude')
ax.scatter(gps_latlon_df.index, gps_latlon_df['GPSLongitude'],␣
↪label='longitude')
plt.xticks(rotation=30)
plt.legend(loc='best')
plt.show()
```



1.5 Load Phone Data

This is data from my old cell phone, a Redmi Note 10 Pro (CN), with the GPSLogger app installed.

It was placed in the tow rope bag on the right hand side of the car. I occasionally pulled it out to check the data but otherwise it remained inside the vehicle at all times.

[9]:
```
phone_gps_df = pd.read_csv('data/20260402.csv', parse_dates=['time'])
phone_gps_df = phone_gps_df.set_index(pd.to_datetime(phone_gps_df['time'],␣
↪utc=True)).drop(columns=['time'])
phone_gps_df = phone_gps_df.rename(columns={'lat': 'GPSLatitude', 'lon':␣
↪'GPSLongitude'})

phone_gps_df.index = phone_gps_df.index.tz_convert('UTC')

phone_gps_df
```

[9]:
```
                                  GPSLatitude  GPSLongitude  elevation  \
time
2026-04-02 15:37:00.705000+00:00    49.198326   -123.141634      -14.3
2026-04-02 15:37:08.255000+00:00    49.198198   -123.142273      -54.1
2026-04-02 15:37:14.051000+00:00    49.198240   -123.142112      -53.8
2026-04-02 15:37:19.774000+00:00    49.198258   -123.142022      -61.8
2026-04-02 15:37:25.449000+00:00    49.198260   -123.141987      -63.4
…                                       …           …        …
2026-04-02 23:15:32+00:00           49.198688   -123.145485      -23.6
2026-04-02 23:15:34+00:00           49.198688   -123.145485      -23.6
2026-04-02 23:15:36+00:00           49.198688   -123.145485      -23.6
2026-04-02 23:15:37+00:00           49.198688   -123.145485      -23.6
2026-04-02 23:15:39+00:00           49.198688   -123.145485      -23.6

                                  accuracy  bearing     speed  satellites  \
time
2026-04-02 15:37:00.705000+00:00    29.894      NaN       NaN           0
2026-04-02 15:37:08.255000+00:00    10.000   301.11  2.398091           7
2026-04-02 15:37:14.051000+00:00    10.600   292.22  0.131254           6
2026-04-02 15:37:19.774000+00:00    10.500   318.50  2.168525           8
2026-04-02 15:37:25.449000+00:00    10.400   339.27  0.180668           8
…                                     …       …        …          …
2026-04-02 23:15:32+00:00            1.300    64.14  0.000000          44
2026-04-02 23:15:34+00:00            1.300    64.14  0.000000          45
2026-04-02 23:15:36+00:00            1.300    64.14  0.000000          45
2026-04-02 23:15:37+00:00            1.300    64.14  0.000000          44
2026-04-02 23:15:39+00:00            1.300    64.14  0.000000          45

                                 provider  hdop  vdop  …  dgpsid  activity  \
time                                                   …
2026-04-02 15:37:00.705000+00:00  network   NaN   NaN  …     NaN       NaN
2026-04-02 15:37:08.255000+00:00      gps  2.20  0.96  …     NaN       NaN
2026-04-02 15:37:14.051000+00:00      gps  1.61  0.93  …     NaN       NaN
2026-04-02 15:37:19.774000+00:00      gps  1.54  0.96  …     NaN       NaN
2026-04-02 15:37:25.449000+00:00      gps  1.22  0.84  …     NaN       NaN
…                                     …    …    …  …     …        …
2026-04-02 23:15:32+00:00             gps  0.39  0.52  …     0.0       NaN
```

```
2026-04-02 23:15:34+00:00            gps  0.38  0.52  …     0.0       NaN
2026-04-02 23:15:36+00:00            gps  0.38  0.52  …     0.0       NaN
2026-04-02 23:15:37+00:00            gps  0.41  0.53  …     0.0       NaN
2026-04-02 23:15:39+00:00            gps  0.38  0.52  …     0.0       NaN

                                 battery  annotation   timestamp_ms  \
time
2026-04-02 15:37:00.705000+00:00      97         NaN  1775144220705
2026-04-02 15:37:08.255000+00:00      97         NaN  1775144228255
2026-04-02 15:37:14.051000+00:00      97         NaN  1775144234051
2026-04-02 15:37:19.774000+00:00      97         NaN  1775144239774
2026-04-02 15:37:25.449000+00:00      97         NaN  1775144245449
…                                   …          …            …
2026-04-02 23:15:32+00:00             80         NaN  1775171732000
2026-04-02 23:15:34+00:00             80         NaN  1775171734000
2026-04-02 23:15:36+00:00             80         NaN  1775171736000
2026-04-02 23:15:37+00:00             80         NaN  1775171737000
2026-04-02 23:15:39+00:00             80         NaN  1775171739000

                                                     time_offset      distance  \
time
2026-04-02 15:37:00.705000+00:00  2026-04-02T08:37:00.705-07:00      0.000000
2026-04-02 15:37:08.255000+00:00  2026-04-02T08:37:08.255-07:00     48.546146
2026-04-02 15:37:14.051000+00:00  2026-04-02T08:37:14.051-07:00     61.173478
2026-04-02 15:37:19.774000+00:00  2026-04-02T08:37:19.774-07:00     68.023230
2026-04-02 15:37:25.449000+00:00  2026-04-02T08:37:25.449-07:00     70.573060
…                                                            …           …
2026-04-02 23:15:32+00:00         2026-04-02T16:15:32.000-07:00  21020.905981
2026-04-02 23:15:34+00:00         2026-04-02T16:15:34.000-07:00  21020.905981
2026-04-02 23:15:36+00:00         2026-04-02T16:15:36.000-07:00  21020.905981
2026-04-02 23:15:37+00:00         2026-04-02T16:15:37.000-07:00  21020.905981
2026-04-02 23:15:39+00:00         2026-04-02T16:15:39.000-07:00  21020.905981

                                 starttimestamp_ms     profile_name  \
time
2026-04-02 15:37:00.705000+00:00      1775144220572  Default Profile
2026-04-02 15:37:08.255000+00:00      1775144220572  Default Profile
2026-04-02 15:37:14.051000+00:00      1775144220572  Default Profile
2026-04-02 15:37:19.774000+00:00      1775144220572  Default Profile
2026-04-02 15:37:25.449000+00:00      1775144220572  Default Profile
…                                             …              …
2026-04-02 23:15:32+00:00             1775144256238  Default Profile
2026-04-02 23:15:34+00:00             1775144256238  Default Profile
2026-04-02 23:15:36+00:00             1775144256238  Default Profile
2026-04-02 23:15:37+00:00             1775144256238  Default Profile
2026-04-02 23:15:39+00:00             1775144256238  Default Profile
```

```
                                   battery_charging
time
2026-04-02 15:37:00.705000+00:00             False
2026-04-02 15:37:08.255000+00:00             False
2026-04-02 15:37:14.051000+00:00             False
2026-04-02 15:37:19.774000+00:00             False
2026-04-02 15:37:25.449000+00:00             False
…                                            …
2026-04-02 23:15:32+00:00                    False
2026-04-02 23:15:34+00:00                    False
2026-04-02 23:15:36+00:00                    False
2026-04-02 23:15:37+00:00                    False
2026-04-02 23:15:39+00:00                    False

[12977 rows x 23 columns]
```

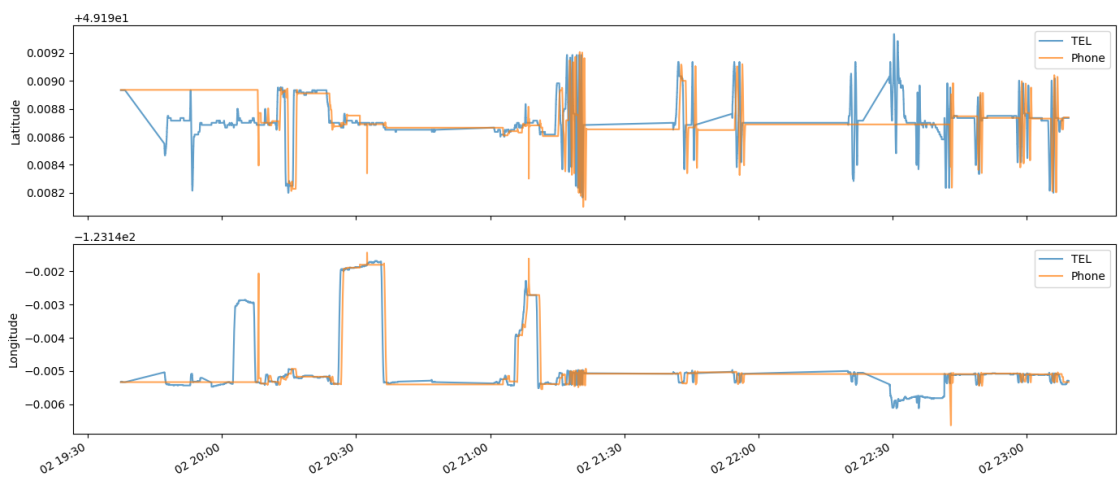
1.6 Compare on Satellite Map

[10]:
```python
import folium

center = [gps_latlon_df['GPSLatitude'].mean(), gps_latlon_df['GPSLongitude'].
 ↪mean()]

m = folium.Map(location=center, zoom_start=18, max_zoom=21, tiles='Esri.
 ↪WorldImagery')

# Plot TEL GPS [RED]
folium.PolyLine(
    locations=gps_latlon_df[['GPSLatitude', 'GPSLongitude']].values.tolist(),
    color='red',
    weight=3,
    opacity=0.9
).add_to(m)

# Plot phone GPS [CYAN]
folium.PolyLine(
    locations=phone_gps_df[['GPSLatitude', 'GPSLongitude']].values.tolist(),
    color='cyan',
    weight=3,
    opacity=0.9
).add_to(m)

m
```

[10]:
```
<folium.folium.Map at 0x19b124dacf0>
```

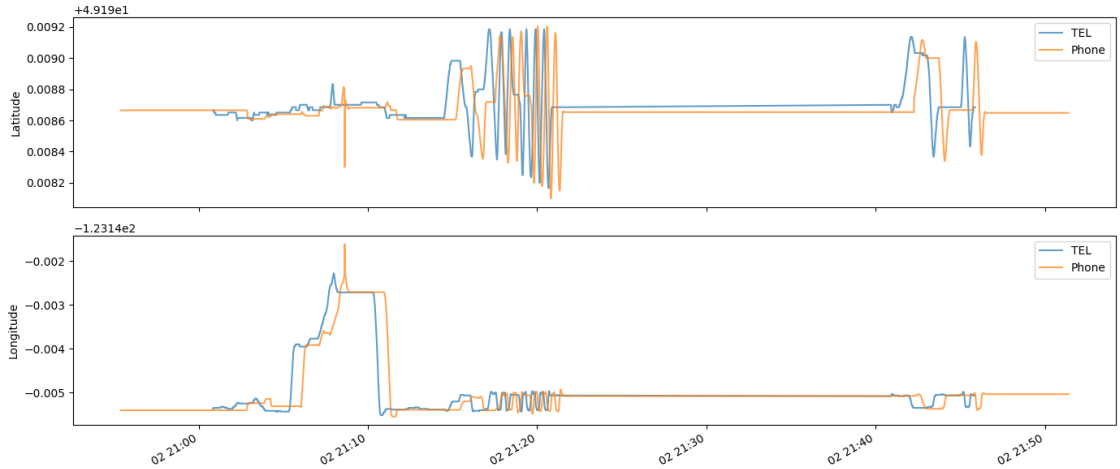
1.7 Compare on Time-Domain

[11]:
```
def compare_phone_tel(start, end):
    fig, axes = plt.subplots(2, 1, figsize=(14, 6), sharex=True)

    axes[0].plot(gps_latlon_df.loc[start:end].index, gps_latlon_df.loc[start:
↪end]['GPSLatitude'], label='TEL', alpha=0.7)
    axes[0].plot(phone_gps_df.loc[start:end].index, phone_gps_df.loc[start:
↪end]['GPSLatitude'], label='Phone', alpha=0.7)
    axes[0].set_ylabel('Latitude')
    axes[0].legend()

    axes[1].plot(gps_latlon_df.loc[start:end].index, gps_latlon_df.loc[start:
↪end]['GPSLongitude'], label='TEL', alpha=0.7)
    axes[1].plot(phone_gps_df.loc[start:end].index, phone_gps_df.loc[start:
↪end]['GPSLongitude'], label='Phone', alpha=0.7)
    axes[1].set_ylabel('Longitude')
    axes[1].legend()

    fig.autofmt_xdate()
    plt.tight_layout()
    plt.show()

start = max(gps_latlon_df.index.min(), phone_gps_df.index.min())
end = min(gps_latlon_df.index.max(), phone_gps_df.index.max())

compare_phone_tel(start, end)
```

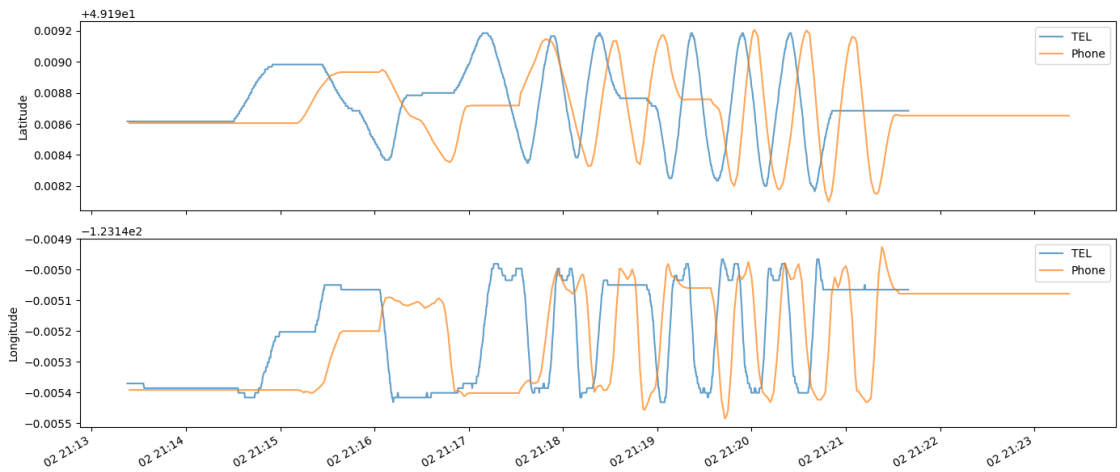


[12]:
```
from datetime import timedelta
```

```
compare_phone_tel(start + timedelta(0, 3600*1.3), end - timedelta(0, 3600*1.3))
```



[13]:
```
sample_start = start + timedelta(0, 3600*1.6)
compare_phone_tel(sample_start, sample_start + timedelta(0, 10*60))
```



There is a notable time shift between the phone's values and the TEL GPS values. We can compute the best fit time shift by maximizing the cross-correlation of the two signals.

1.8 Find & Correct the Time Shift

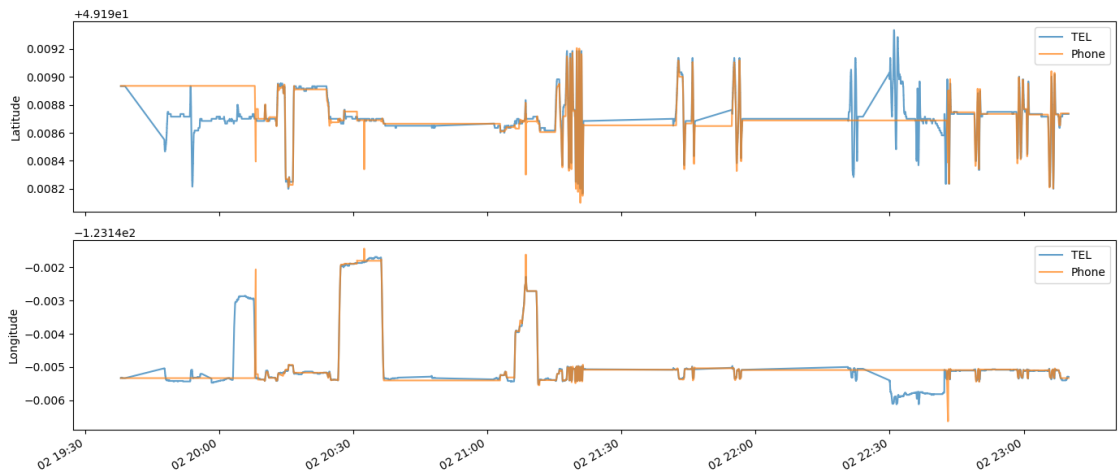
[14]:
```
import numpy as np
from scipy.interpolate import interp1d

# Common time grid at 1-second resolution
```

```
t_start = max(gps_latlon_df.index.min(), phone_gps_df.index.min())
t_end = min(gps_latlon_df.index.max(), phone_gps_df.index.max())
t_grid = pd.date_range(t_start, t_end, freq='1s')
t_seconds = np.arange(len(t_grid))

# Interpolate both onto the common grid using latitude
def interpolate_to_grid(df, col, t_grid):
    t_ns = df.index.astype(np.int64)
    t_grid_ns = t_grid.astype(np.int64)
    f = interp1d(t_ns, df[col].values, bounds_error=False, fill_value=np.nan)
    return f(t_grid_ns)

tel_lat = interpolate_to_grid(gps_latlon_df, 'GPSLatitude', t_grid)
phone_lat = interpolate_to_grid(phone_gps_df, 'GPSLatitude', t_grid)

# Mask NaNs
mask = ~np.isnan(tel_lat) & ~np.isnan(phone_lat)
a = tel_lat[mask] - tel_lat[mask].mean()
b = phone_lat[mask] - phone_lat[mask].mean()

# Cross-correlation
corr = np.correlate(a, b, mode='full')
lag = np.argmax(corr) - (len(b) - 1)

print(f"Time shift: {lag} seconds (positive = TEL is ahead of Phone)")
```

```
Time shift: -39 seconds (positive = TEL is ahead of Phone)
```

[15]:
```
gps_latlon_df_aligned = gps_latlon_df.copy()
gps_latlon_df_aligned.index = gps_latlon_df_aligned.index - pd.
 ↪Timedelta(seconds=lag)

start = max(gps_latlon_df_aligned.index.min(), phone_gps_df.index.min())
end = min(gps_latlon_df_aligned.index.max(), phone_gps_df.index.max())

fig, axes = plt.subplots(2, 1, figsize=(14, 6), sharex=True)

axes[0].plot(gps_latlon_df_aligned.loc[start:end].index, gps_latlon_df_aligned.
 ↪loc[start:end]['GPSLatitude'], label='TEL', alpha=0.7)
axes[0].plot(phone_gps_df.loc[start:end].index, phone_gps_df.loc[start:
 ↪end]['GPSLatitude'], label='Phone', alpha=0.7)
axes[0].set_ylabel('Latitude')
axes[0].legend()

axes[1].plot(gps_latlon_df_aligned.loc[start:end].index, gps_latlon_df_aligned.
 ↪loc[start:end]['GPSLongitude'], label='TEL', alpha=0.7)
```

```
axes[1].plot(phone_gps_df.loc[start:end].index, phone_gps_df.loc[start:
 ↪end]['GPSLongitude'], label='Phone', alpha=0.7)
axes[1].set_ylabel('Longitude')
axes[1].legend()

fig.autofmt_xdate()
plt.tight_layout()
plt.show()
```



[ ]:

```
```